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Genomic rearrangements analysis of human genome

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Abstract

Genomic rearrangements describe loss or gain of chromosomal material of the size ranging from a couple of hundred base pairs and generally they are mediated by NAHR mechanism. They are reason of formation of most common and better-delineated genomic disorders. To better understand their mechanisms in human genome we selected those which are located in potential regulatory regions of haploinsufient genes verified experimentally and inferred computationally [Hurles *et al.*, PLoS Genet., vol. 458]. Thus, potential regulatory regions were cross-referenced with the chromosomal microarray analysis (CMA) database from the Medical Genetics Laboratories (MGL) of Baylor College of Medicine (BCM) of 27,633 patients. Our study finds dependences between genome architectural features and genomic rearrangements within these potential regulatory regions.

Słowa kluczowe

position-effect, haploinsufficiency, long noncoding RNA

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Tytuł pracy w języku angielskim

Genomic rearrangements analysis of human genome

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Chapter 1

Introduction

The main aim of this thesis is better understanding mechanisms and phenotypic effects of genomic rearrangements in human genome.

Chapter 1 presents the biological knowledge required to get the full understanding of this work. Especially, it describes a position effect mechanism and contains a definition of haploinsufficiency. Many questions in genetics still remain unsolved, such as about the origin of copy number variations (CNVs), their evolutionary adaptation and impact in many human genetic disorders. Tools such as array comparative genomic hybridization allow scientists to look at copy-number variations in detail and examine their significance under position effect, haploinsufficiency mechanisms and in general.

Chapter 2 presents materials and our methodology to discover the role of higher-order genomic architectural features around position effect and haploinsufficient genes. To detect dependences between these features and CNVs we calculated Pearson and Spearman correlation coefficients.

Presentation of our workflow and an illustrative example of one of the potential regulatory genomic region from UCSC browser with marked crucial genomic features are concerned in chapter 3. Moreover, we visualizated previously mentioned Pearson and Spearman correlation coefficients as correlation matrixes.

1.1. Genome organization and long-range control

Human genome is usually classified into coding and noncoding DNA sequences. DNA fragment that can be transcribed into mRNA and then translated into a protein is called a coding sequence. Noncoding DNA is defined as all of the DNA sequences that do not code proteins. The vast majority of 3.3 billion nucleotides in the human genome are noncoding sequences (nearly 99%) and coding ones are only a small fraction of the genome (circa 1%) [26]. Initially, noncoding DNA was named as "junk DNA" due to its unknown biological function. However comparative genomics studies uncover that the majority of mammalian-conserved regions consist of noncoding elements. This indicates that functional elements with regulatory and other functions are situated in noncoding regions of the human genome [21].

There are many classes of noncoding DNA: pseudogenes, introns and untranslated regions of mRNA, genes for noncoding RNA (ncRNA), regulatory DNA sequences, repetitive DNA sequences and transposable genetic elements.

Pseudogenes are inactive copies of protein-coding genes.

An intron is a noncoding part of the gene's nucleotide sequence, it separates the coding parts of the gene (exons). It is removed from mRNA during mRNA maturation (RNA

splicing) and then final product of a gene can be generated.

Genes for noncoding RNA are genes that code RNA which won't be translated into proteins. Bioinformatic studies suggest the existence of thousands of them in the human genome. It is known that ncRNA rearragments can cause a variety of diseases such as autism [9] or Alzheimer's disease [14]. A particular subclass of the ncRNAs are long non-coding RNAs (long ncRNAs, lncRNA) - ncRNAs that are longer than 200 nucleotides. Recently, they have been discovered in imprinting and long-range gene regulation with properties of enhacers [38].

Approximately half of the human genome are repetetive DNA sequences. They are patterns that occur in multiple copies throughout the genome. The copies can be directly adjacent to each other (tandem repeats), lie on both ends of a sequence (terminal repeats) or can have ability to change its position within the genome, replicate and insert copies of themselves at other locations (called interspersed repeats, transposable genetic elements or transposons). There are two classes of transposons: retrotransposons and DNA transposons.

Many genes require not only coding sequence but also correctly functioning regulatory sequences, that is multiple cis-acting genomic regions (regions on the same chromosome) that are required for correct spatial, temporal and quantitative expression of them [24]. It is accomplished through the sequence-specific binding of proteins (transcription factors) which may promote (activators) or blocking (repressors) transcription of genetic information from DNA to mRNA. They can bind to the DNA sequence located upstream, downstrem of transcription unit or even within introns of neighboring genes (see Figure 1.1).

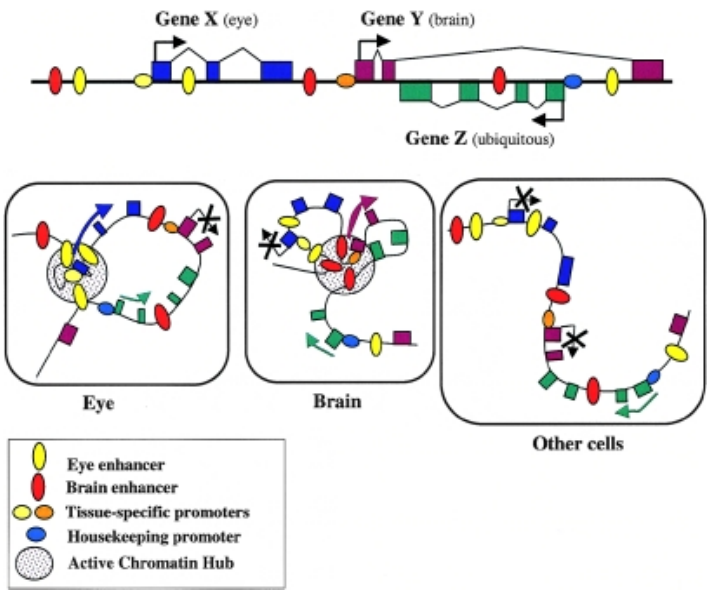

Figure 1.1: Schematic representation of a theoretical gene loci. This schematic region contains two tissue-specific genes and one housekeeping gene. Gene X (with blue exons) is expressed in eye tissue, gene Y (purple) in brain tissue and gene Z (green) is all kinds of tissues. Gene expressions depends on the appropriate formation of the chromatin (Active Chromatin Hub, ACH) allowing to interact various tissue-specific cis-acting elements, such as enhancers, with corresponding gene promoters. Activity of the housekeeping gene (green) does not rely on tissue-specific ACH formation. Functional gene expression modules can physically overlap (gene Y and gene Z). In other words, cis-regulatory elements that influence the expression of one gene can be located within introns of another, adjacent gene, or can in some cases even be located beyond the neighbouring gene. The figure from [22].

1.2. Position effect

Correct gene expression requires the presence of its complete coding sequence as well as properly functioning regulatory control. It can be disrupted by: mutation in the gene or physical separation of the gene from its full set of regulatory elements, alteration of local or global regulation of chromatin structure within gene or its regulatory elements [22].

Traditionally, the term 'position effect' was used to describe situations where level of a gene expression was altered by genes chromosomal enviroment, while the gene was intact [23]. Dirk A. Kleinjan *et al.* [23] suggest replacing the term 'position effect mechanism' with 'cis-ruption' to describe general causes of genetic disorders caused by dysfunction of the cis-regulatory elements of the disease gene locus.

In recent years, an increasing number of such disease-related position-effect cases have come to light and examples fo them are reviewed in [22].

1.3. Haploinsufficiency

Haploinsufficiency (HI) is a situation when in diploid organism a single functional copy of a gene is not enough to prevent a disease (the other copy is disactivated by mutation). In this case one copy does not produce enough of gene product (usually a protein).

Over the past 20 years over 300 such genes have been experimentally identified in human genome. However there are also haploinsufficient genes predicted computationally. Matthew E. Hurles et.al [10], by comparing known HI genes with available selected properties with genes where loss of only one copy does not result in disease, discovered over 12000 putative HI genes. Their predictive HI model is based on genomic (i.e. number of protein domains), evolutionary (i.e. identity with closest paralog), functional (i.e. embryonic expression), protein-protein (i.e. protein-protein interaction proximity to HI), and genetic interaction (i.e. gene network proximity degree to HI) properties and similarities between these two groups of genes.

Haploinsufficiency is often caused by copy-number variation (CNV). One example is the neurodevelopmental disorder called Williams syndrome caused by the haploinsufficiency of genes situated on chromosome 7 (7q11.23). In this case 28 genes are removed from genome as a result of 1.6 Mb deletion.

1.4. Copy-number variation (CNV)

Copy-number variation affects haploinsufficient genes, genes critical for normal development or maintenance results in human genetic diseases.

1.4.1. CNV definition

The term "copy-number variation" (CNV) refers to an alteration of the DNA that results in having an abnormal number of one or more sections of the DNA. In contrast to single-nucleotide polymorphism (SNP), which affects only one single nucleotide base, it corresponds to regions encompassing from a couple of hundreds base pairs to even 5 megabases (the bigger ones are visible microscopically). It mainfests as sequences deletions (loss of genetic material) or duplications (gain of it) on certain chromosomes. Together with inversions and translocations they change structure of chromosomes and therefore they are classified as structural chromosomal rearagments [12, 8].

1.4.2. CNV sources

Usually they occur with a breakage in the DNA double helices at two different locations followed by reconstruction in an abnormal combination of the broken ends. Unless such double-strand breaks in a cell are repaired, they are potentially lethal. There are several proposed major repair systems and mechanisms underlying CNVs in the human genome: nonhomologous end-joining (NHEJ) [35], microhomology-mediated end joining (MMEJ) [35], nonallelic homologous recombination (NAHR) [22], stalling and template switching / microhomology-mediated break-induced replication (FoSTeS/MMBIR) [42], LINE-1 retrotransposition [28].

Generally most of them are caused by nonallelic homologous recombination (NAHR) mediated by repetitive sequences mostly such as low-copy repeats (LCRs, also called segmental duplications, SD) or rarely retrotranspons like Alu [29] or those belonging to long interspersed element (LINE) family [33]. LCRs are highly homologous sequence elements (95% in identity), typically 10–300 kb in length. Most of LCRs tend to be concentrated in "hotspots". In comparision to allelic homologous recombination (AHR), which is a normal process that is a part of meiosis occurring between sister chromatids and is important an aspect of the generation of genetic variation in species, during NAHR sequences are recombined on different chromosomes, different positions of homologous chromosomes or even two sequences on the same chromosome [28, 8] (see Figure 1.2). Some evidence has suggested that NAHR frequency correlates with LCR length and is inversely proportional to the distance between the LCRs [28]. They can either be caused by de novo mutations or be inherited [11]. The same pairs of LCRs can mediate both mitotic and meiotic NAHR events [28].

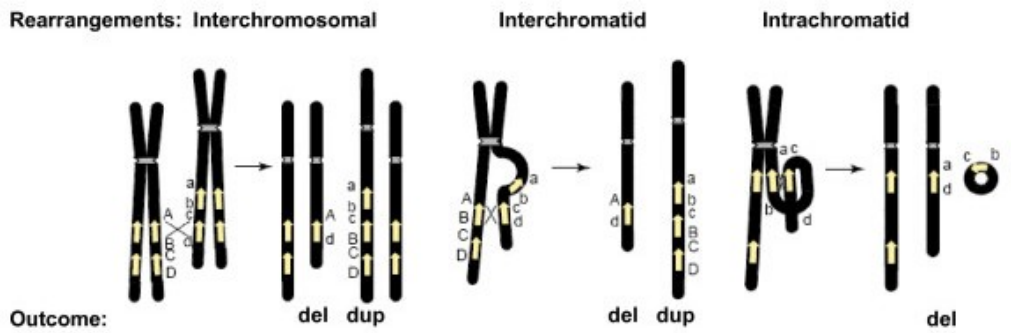

Figure 1.2: Schematic example of representation low-copy repeat/nonallelic homologous recombination (LCR/NAHR)-based mechanisms for CNVs. LCRs are represented as arrows in yellow with the orientation indicated by the arrowhead. Capital letters refer to flanking unique sequences and lower-case letters to the homologues ones on the other strand. The figure depicts interchromosomal, interchromatid and intrachromatid NAHR using LCR pairs in direct orientation. However LCRs pairs can be arranged also as inverted and complex LCRs [37, 28]. Figure adapted from [28].

Other major mechanism is called nonhomologous end-joining (NHEJ), which repair DNA double-strand breaks by simple ligating their two ends together [28] (see Figure 1.3 (a)). Here, in comparision to other mechanisms the microhomology is not required. NHEJ is referred to as "non-homologous" because the break ends are directly ligated without the need for a homologous template. It has been described in organisms ranging from bacteria to mammals.

Last proposed mechanism is a DNA replication–based mechanism of fork stalling and template switching (FoSTeS). It occurs mitotically and may be a major mechanism for duplications formation (see Figure 1.3 (b)). DNA replication is a process of producing two identical

replicas from one original DNA molecule occuring in all living organisms. According to the FoSTeS, during rare replication defects, one fork from broken DNA invades a fork from another region on the same chromosome. It results in a template switch between these DNA regions. The FoSTeS mechanism has been further generalized to microhomology-mediated break-induced replication (MMBIR) [42].

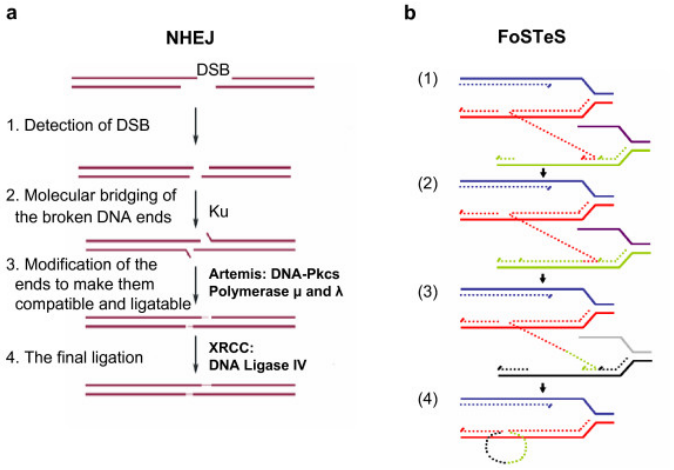

Figure 1.3: (a) Representation of steps of non-homologous end-joining (NHEJ). When occur a double-stranded DNA break (DSB) series of proteins prepare and then ligate two ends of broken DNA strands. During preparation takes place the cooperative binding and phosphorylation of artemis by the DNAPkcs and Polymerase μ and λ to make broken ends of DNA compatibe and ligatible. In final ligation step there is involved DNA ligase IV using XRCC4 protein. (b) Stalling and template switching mechanism (FoSTeS). The figure from [28].

Rearrangements of the human genome can be also categorized into two groups based on breakpoint analysis: recurrent and nonrecurrent rearagments. The first share a common rearagment interval and size. Nonrecurrent rearrangements have variable breakpoints (see Figure 1.4). Most of the recurrent rearrangements result from NAHR between LCRs and NHEJ. FoSTes is the proposed mechanism to explain some nonrecurrent rearrangements [42].

1.4.3. Role in diseases

CNV characterization is important for both the basic understanding of many diseases and their diagnoses. Abnormal phenotypes caused by deletions are generally more severe than those caused by duplications. Deletions visible by microscope occur approximately in 1 in every 7000 births. The better-characterized deletion disorders are cri du chat syndrome and Prader-Willi syndrome. Both of them are characterized by mental retardation, as well as a number of physical defects [8]. Examples of diseases caused by microduplication are Potocki-Lupski syndrome with symptomes including mild mental retardation and autism [31] and Charcot–Marie–Tooth disease characterised by progressive loss of muscle tissue and touch sensation across various parts of the body [32].

1.5. Array comparative genomic hybridization (aCGH)

Copy number variations can be detected by various techniques. An ideal one should precisely quantify the copy numbers and mark their breakpoints across the whole genome. So far there

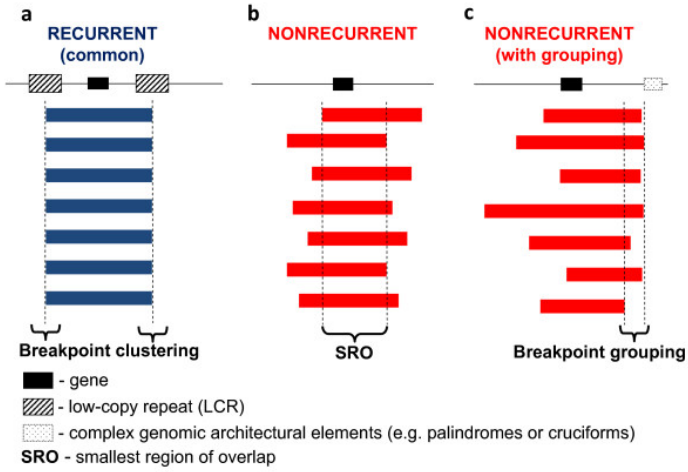

Figure 1.4: Recurrent and nonrecurrent genomic rearragments. The long thin line represents genome sequence. The black rectangle depicts a gene and blue and red bars rearrangements (duplications, deletions or inversions) with their breakpoints. (a) Recurrent rearangements are flanked by LCR sequences and breakpoints are clustered inside the LCRs. (b) Non-recurrent ones are scattered in different directions. They share a common genomic region of overlap. The smallest overlapped region include a region (here gene) necessary for phenotype identification. (c) Non-recurrent with grouping of one breakpoint localized in small genomic region near complex genomic elements, e.g. palindroms or cruciforms. The figure from [28].

are several techniques including fluorescent in situ hybridization (FISH) [36], comparative genomic hybridization (CGH) [20], array comparative genomic hybridization (array CGH, aCGH) [13], karyotyping with single nucleotide polymorphism (SNP) array [7] and next-generation sequencing (NGS) [27]. NGS has the potential to offer the highest resolution but it has higher prices owing to the greater sequencing depth required. In our work we used data from aCGH approach and therefore we will focus on this technique below.

Array CGH is a technology that allows detection of submicroscopic chromosome imbalances such as microdeletions and microduplications. Steps of methodology for aCGH analysis are shown in Figures 1.5. In general it compares a test DNA with a control (reference) DNA sample and identifies differences between these two sets of DNA. Test DNA can be extracted from i.e. blood, skin or fetal cells. Both of then are labelled with a different coloured fluorescent dye, mixed together and applied to a microarray. Next, a digital imaging system is used to quantify the ratio between fluorescence intensities of the labeled DNA probes that have hybridized to each target. It inform about relative copy number of sequences in the test genome which is compared to the normal genome.

The advantages of aCGH are that it can detect genomic rearangements of any locus represented on an array and it can quickly analyze hundreds or thousands of discrete regions of the whole genome (it gives a global view of chromosomes). It offer higher resolution than traditional cytogenetic methods.

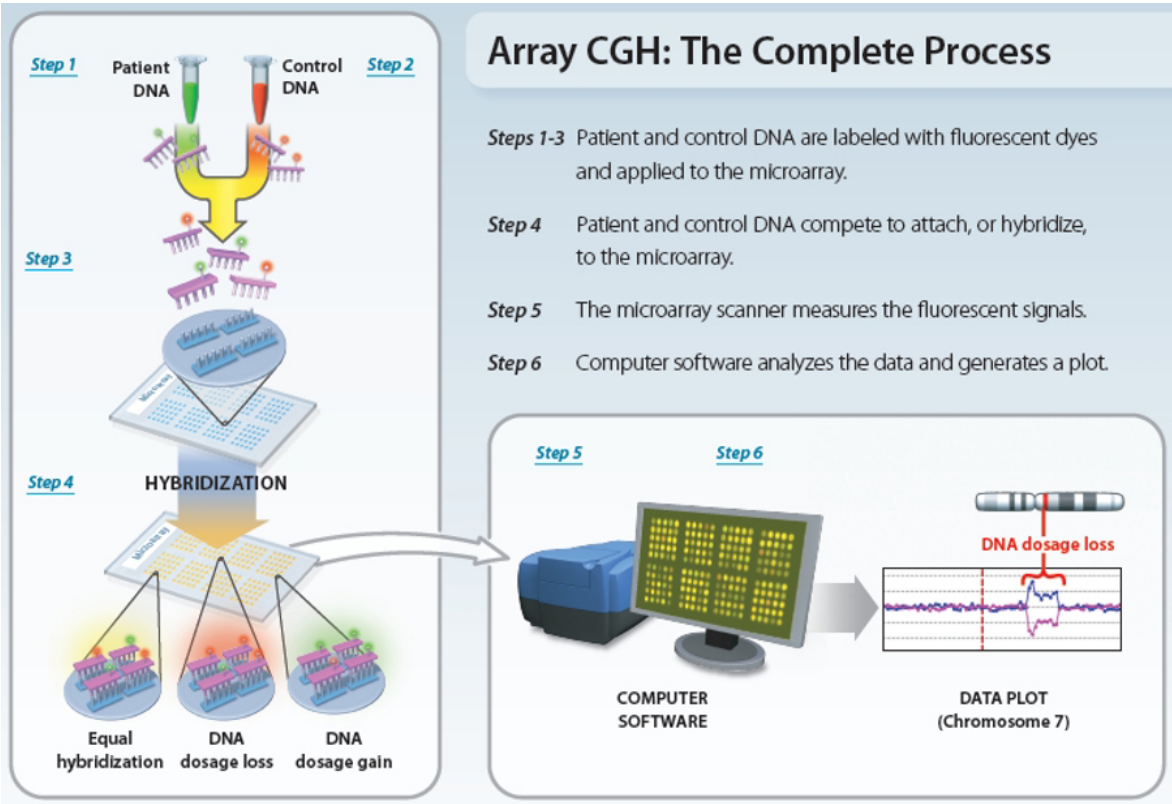

Figure 1.5: Diagram of the microarray-based comparative genomic hybridisation (aCGH) procedure. The figure from [39].

Chapter 2

Materials and methods

2.1. Selected regions prone to genomic instability

We characterized potential NAHR-prone genomic regions. They are located both upstream and/or downstream of selected experimentally proven effect and haploinsufficient (HI) genes (listed in Table 2.1). In addition, we took surrounding regions of computionaly predicted HI genes from [10] (Dataset 1). Particularly, we set their distance from 3' site to average 3' distance of genes specified in Table 2.1 (equal to 365 bp), respectively for 5' site distance (481 bp) and the average length of these regulatory regions was $146,980$ bp.

2.2. CMA database analysis/Genomic copy-number analysis

To investigate these genomic regions, we characterized submicroscopic chromosome deletions and duplications intersecting with them.

Genomic rearrangements were identified by aCGH technique and derived from the chromosomal microarray analysis (CMA) database of 27,633 patients subjected to genome-wide assay. The CMA database comes from the Medical Genetics Laboratories (MGL) of Baylor College of Medicine (BCM).

Firstly, we filtered rearrangements from the aCGH data leaving only the changes that are marked both as "exported" and "loss/gain". In total there are 1580 such deletions and 1671 duplications for selected experimentally HI genes, and 194667 and 263257 respectively for computationally predicted HI genes. Then we selected gene free rearrangements (i.e. with only lncRNA genes). Based on gene databases available from UCSC MySQL database [6]: UCSC Known Genes Based on UniProt, RefSeq, GenBank mRNA and Ensembl Genes we selected deletions and duplications in which noncoding genes of length more than 200 bp can be located. Visualizations of the number of these rearrangements within selected genomic regions are shown in Figure 2.1. Each row in histograms represents the regulatory region and on the X-axis counts of deletions/duplications of length less or equal to 1 Mb, 1–1.5 Mb, 1.5–2 Mb, 2–2.5 Mb are demonstrated.

Gene	3' or 5' side	Distance from a gene [kb]	Deletion/duplication size [kb]	Distance from translocation/inversion	Location
ATOH7	5'	20	6.523		chr10:70,007,124-70,013,647
BMP2	3'	110	5.5		chr20:6,860,500-6,866,000
CDK6	5'			66	chr7:92,529,212-92,692,212
DLX5	3'			1000	chr7:95,784,517-96,054,143
FOXC1	5'			25	chr6:1,450,681-1,589,127
FOXC2	3'			120	chr16:86,722,447-86,823,447
FOXF1	5'	250	70		chr16:86,215,000-86,287,054
FOXL2	5'	283	7.4		chr3:138,949,151-138,956,509
GLI3	5'	40	80		chr7:42,316,618-42,356,618
HBA1	3'	18			chr16:245,519-250,519
HBA2	3'	30			chr16:253,708-258,708
HBB	5'	50			chr11:5,300,472-5,303,435
HBE1	5'	9,5			chr11:5,301,137-5,302,803
IRF6	3'	1900			chr1:209,989,128-209,989,367
IRGM	5'	2,7	20		chr5:150,205,085-150,225,529
LCT	5'	14	0,104		chr2:136,608,189-136,609,228
NR0B1 (DAX1)	5'	11	257		chrX:30,338,495-30,595,495
PAX6	3'	11			chr11:31,494,341-31,795,341
PITX2	5'			15	chr4:111,559,254-112,031,254
PLP1	3'	136	150		chrX:103,183,545-103,333,545
PLP1	5'			70	chrX:102,933,754-102,961,754
PMP22	5'	33	186		chr17:15,197,093-15,383,093
POU3F4	5'	900	8		[1]
PTCHD1	5'	150-370			chrX:23,010,079-23,290,079
REEP3	3'			43	chr10:65,424,971-65,487,971
RET	intron 1				chr10:43,572,517-43,595,517
RUNX2	5'			829	chr6:44,417,054-44,467,054
SALL1	5'			180	chr16:51,365,183-52,043,183
SDC2	3'			30	chr8:97,631,037-97,666,499
SHH	5'			15	chr7:155,619,967-156,473,570
SHH	5'	1000			chr7:156,582,734-156,585,733 [2]
SHOX	3'	150	100		chrX:757,558-857,558
SIX3	5'			200	chr2:44,769,037-44,969,037

Continued on next page

[1] chrX:81,787,344-81,789,344; chrX:81,840,344-81,842,344; chrX:82,591,077-82,592,410; chrX:82,750,477-82,751,810; chrX:82,758,743-82,761,744; chrX:81,839,105-81,847,188

[2] intron 5 of LMBR1

Table 2.1 – *Continued from previous page*

Gene	3' or 5' side	Distance from a gene [kb]	Deletion/duplication size [kb]	Distance from translocation/inversion	Location
SOST	3'	35	52		chr17:41,749,099-41,801,156
SOX3	3'	67	25		chrX:139,495,152-139,520,225
SOX9	3'	1560	36		chr17:71,683,161-71,718,560
SOX9	5'	600	178		chr17:68,217,561-69,737,160
SRY	3'	3			chrY:2,649,520-2,654,896
SRY	5'	1,7			chrY:2,657,681-2,658,569
TBX5	3'			20	chr12:114,595,735-114,771,735
TGFB2	3'			500	chr1:219,117,959-219,257,959
TWIST1	3'			260	chr7:18,734,093-19,150,093

2.3. Genomic features

In the analysis of selected potential regions prone to genomic changes we took into consideration numerous genomic features, which may contribute to long-range transcriptional genes regulation: mammalian conservation, CpG islands, human expressed sequence tag (ESTs) areas, interspersed repeats and low complexity areas, open chromatin, histone modifications, transcription factor dimers binding site, polymerase, enhancers and promoters binding site predictions.

2.3.1. Ensembl database (via Nencki database)

Information about open chromatin, simple genomic features, histone modifications, polymerase, enhancers, promoters binding site predictions are taken from public Nencki MySQL database [25] (the 1_1_hsap_public schema). The main source of this database is the Ensembl funcgen and core database [17] which are describing enrichment of various regulatory features [1]. Search Region contains the promoters' predictions imported into Ensembl funcgen from the cisRED database [34], Open chromatin - result of DNase1 hypersensitivity and FAIRE experiments, and Polymerase, Enhancer binding data from ChIP-Seq experiments from the Ensembl database. Transcription Factor track describes genome-wide scan for transcription factor binding sites motifs with the program matrix-scan (it is a part of the RSAT package [40]), using the public Jaspar, and the commercial Transfac Professional (2013.1) [41].

2.3.2. UCSC database

All other genomic features, i.e. mammalian conservation, CpG islands, human expressed sequence tag (ESTs) areas, lincRNAs transcripts interspersed repeats and low complexity areas, and open chromatin are taken from public UCSC database. Mammalian conservation is predicted by the Multiz and the PhyloP software [4]. Interspersed repeats and low complexity binding by RepeatMasker [5]. Human EST's are derived from the GenBank database [2], LincRNAs transcripts from the Human Body Map [3]. Open chromatin information is from ENCODE database by DNaseI, FAIRE an ChIP experiments from Duke Univeristy, The University of North Carolina at Chapel Hill, and University of Washington on 3 types of cell lines: GM12878, H1-hESC, K562.

The UCSC database the same set of data available on the public UCSC Genome Browser site and both are hosted by the University of California. I has a diverse collection of annotation datasets including mRNA alignments, mappings of DNA repeat elements, gene predictions, gene-expression data and disease-association data.

The UCSC database contains also some information from Regulation, Expression and Variation track belonging to The Encyclopedia of DNA Elements (ENCODE) Consortium and launched by The National Human Genome Research Institute (NHGRI). It holds information about functional elements in the human genome.

2.3.3. TACO

Coverage of motifs of transcription factor dimers were predicted by TACO software. The input data were whole-genome open chromatin profiles data from DNase-seq experiments performed at the University of Washington from The ENCODE Project Consortium [15] and motifs from the Jaspar database [30]. We took motifs with p-value less than 0.05.

TACO (Transcription factor Association from Complex Overrepresentation) is a program to predict cooperative binding cell-type–specific transcription factor dimers to DNA in any

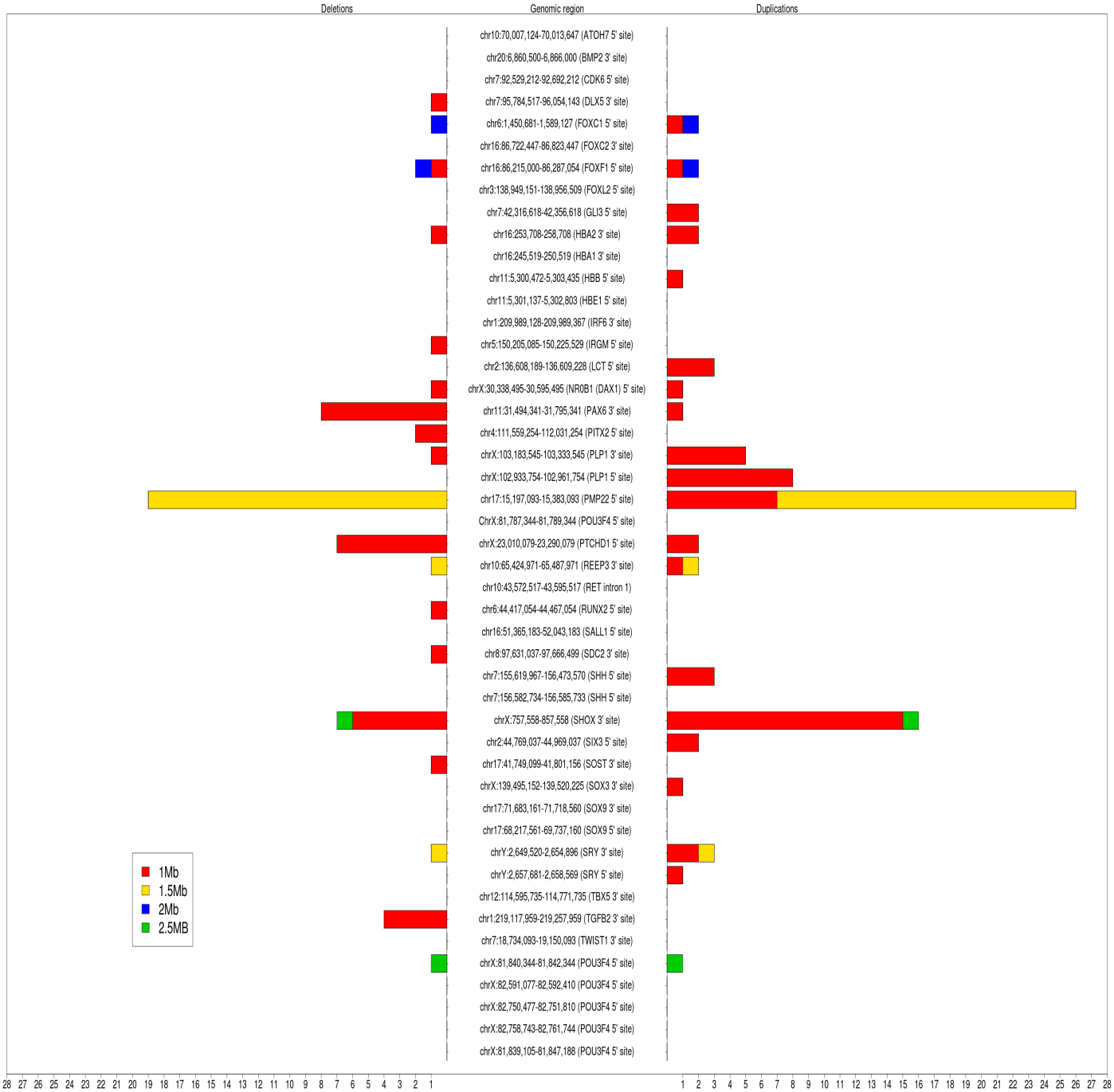

Figure 2.1: Histograms describing counts of deletions/duplications corresponding to the surrounding genomic regions of the position-effect genes. Gene gree deletions/duplciations less or equal to 1 Mb and 1–1.5 Mb, 1.5–2 Mb, 2–2.5 Mb are marked in red, green, blue and purple.

genome-wide set of regulatory regions [19]. A characteristic feature of many known TF dimers is that their motif complexes consist of overlapping motifs. In contrast to other methods this tool takes this aspect into account. Motif complexes are calculated using the algorithm described in [18].

2.4. Correlation

We computed Pearson and Spearman correlation between coverage of deletions and duplications within potential regulatory regions of selected position effect/haploinsufficient genes with coverage or numeric score of selected genomic features.

Coverage is defined as the number of instances of an object in each nucleotide. Genomics features with counted coverage: lincRNAs transcripts from the The Human Body Map catalog [3], human expressed sequence tag (ESTs) from the GenBank database [2], interspersed repeats and low complexity binding prediction by RepeatMasker [5], open chromatin, simple genomic features, histone modifications, polymerase, enhancers, promoters binding site predictions from the Ensembl funcgen database [17] through Nencki database [25].

Features with numeric score are: conservation by phyloP [4] (sumData score), ENCODE [15] Open Chromatin by DNaseI HS and FAIRE experiments from Duke University, University of Washington and UNC Chapel Hill (signalValue/pValue score), CpG islands (obsExp score) from UCSC database [6].

Correlation is a statistical relationship between two random variables or two sets of data. This dependance of two variables means that in some cases value of one of them can be better predicted due to knowledge of the second one than without it. Correlation coefficient measures the degree of correlation two or more variables, specifies the extend interdependence of them. There are several correlation coefficients. Most of them take value between -1 and 1. -1 (negative correlation, decreasing relationship), 0 (lack of correlation), 1 (positive correlation).

The most common of these is the Pearson correlation coefficient. It detects a linear dependance. It is defined as:

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

where cov is the covariance of the two variables, σ_X is the standard deviations of X, μ_X is the mean of X and E is the expectation.

More robust is a Spearman correlation coefficient. It is more sensitive to nonlinear relationship and due to ranked, it is not sensitive to outliers. It is defined as Pearson corellation between the ranked variables:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

where $d_i = x_i - y_i \quad (i = 1, 2, ..., n)$ is a difference between ranks and n number of tested samples. Spearman calculation steps are: descending or ascending ordering variables and then giving variables consecutive natural numbers (ranks) of their ordered positions (i.e. 1,2,3,...).

Chapter 3

Results and discussion

3.1. Overview

The structure of the thesis is outlined in Figure 3.1. We adopted a two-pronged analysis to investigate genomic instability: we took dataset with potential regulatory regions of 1) selected known position effect and haploinsufficient genes (Table 2.1) and 2) haploinsufficient genes predicted computationally by Matthew E. Hurles *et al.* [16]. Then we crosslinked them with genomic rearangements from the MGL BCM CMA database. We counted Spearman and Pearson correlation between coverage of these rearagments and selected genomic features described in *Materials and Methods*.

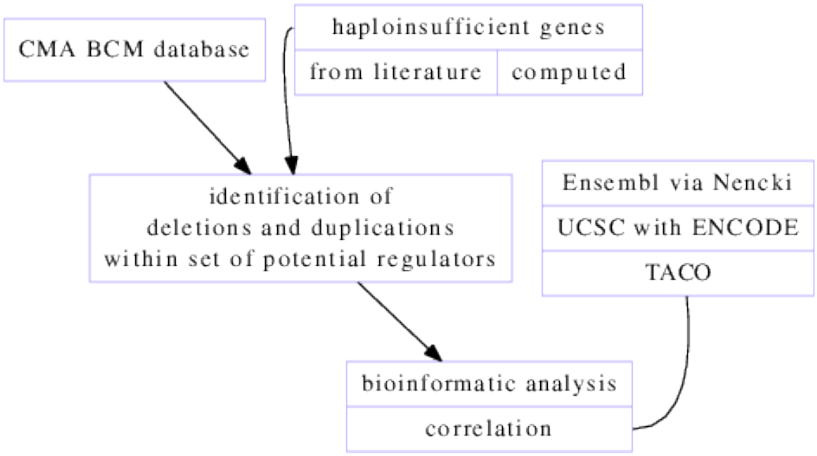

Figure 3.1: Schematic diagram of the workflow.

3.2. An illustrative example

The illustration of UCSC Genome Browser of one of promising potential regulatory regions is shown in Figure 3.2. It represents region near 3' site of human gene TGFB2 (chr1:219,117,959-219,257,959). This region harbors evolutionarily conserved among land vertebrates DNA segments and the fetal lncRNA genes expressed in various tissues and also deletions within it in three patients (KCL165500, KCL165501, KCL167482).

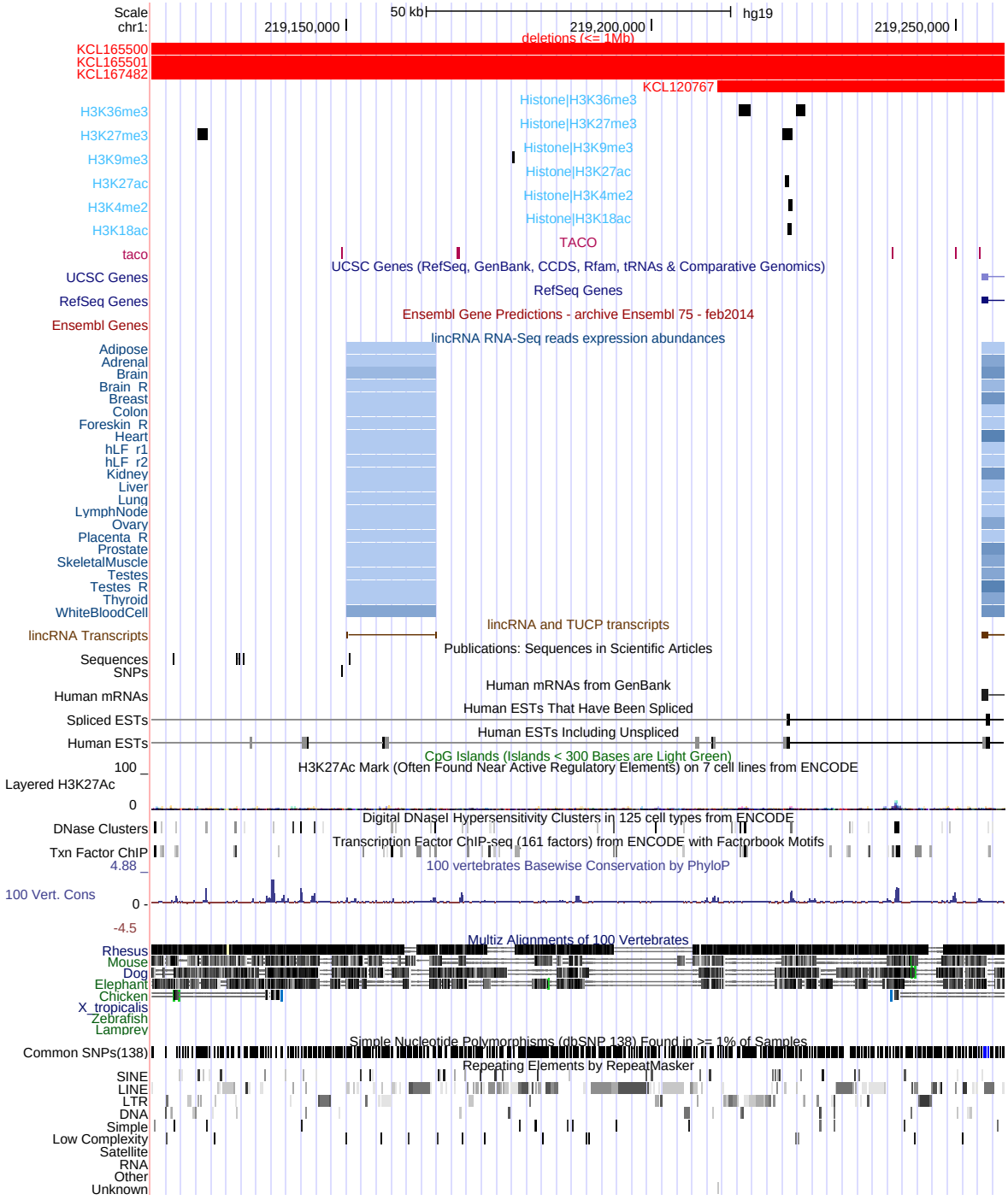

Figure 3.2: Characterization of region chr1:219,117,959-219,257,959 which shows deletions in three patients (KCL165500, KCL165501, KCL167482) and regulatory features: differential histone methylation and acetylation, transcription factor dimers binding sites, level of expression lncRNA genes in various tissues, human EST's, mammalian evolutionary conservation, single nucleotide polimorfisms, interspersed repeats and low complexity binding sites. The figure is drawn according to the 2009 Human Reference Sequence (GRCh37/hg19).

3.3. Correlation

We computed Pearson and Spearman correlation between coverage of deletions and duplications within potential regulatory regions of selected HI genes from literature and predictive

HI model with a level or coverage of genomic features. These are:

- mammalian conservation by phyloP (Conservation),
- cytosine and guanine methylation (CpG islands), CpG islands on All Sequences (in whole genome),
- lincRNAs transcripts from the The Human Body Map catalog [3],
- interspersed repeats and low complexity binding prediction by RepeatMasker (RepeatMasker [5]),
- human ESTs from the GenBank database [2],
- open chromatin data from the Univeristy of Washington (UW DNaseI HS cell-type) and Duke University (Duke DNaseI HS cell-type),
- DNaseI Hypersensitivity Uniform Peaks from ENCODE [15] (DNaseI HS cell-type),
- DNaseI/FAIRE/ChIP Synthesis from ENCODE (Open Chrom Synth cell-type), where cell-types are Gm12878, H1-hesc, K562.
- Open Chromatin, Histone, Transcription Factor, Polymerase, Simple Feature, Enhancer, Search Region from the Ensembl funcgen database through the Nencki database [25].

We generated correlation histograms for all of mentioned ranges of deletions/duplications, but the most informative are those less or equal to 1 Mb. They are shown in Figures 3.3, 3.4, 3.5, 3.6 (potential regulatory regions of selected known HI genes) and 3.3, 3.4, 3.5, 3.6 (potential regulatory regions of HI genes computationally predicted). Every histogram has dendrograms in both axes added.

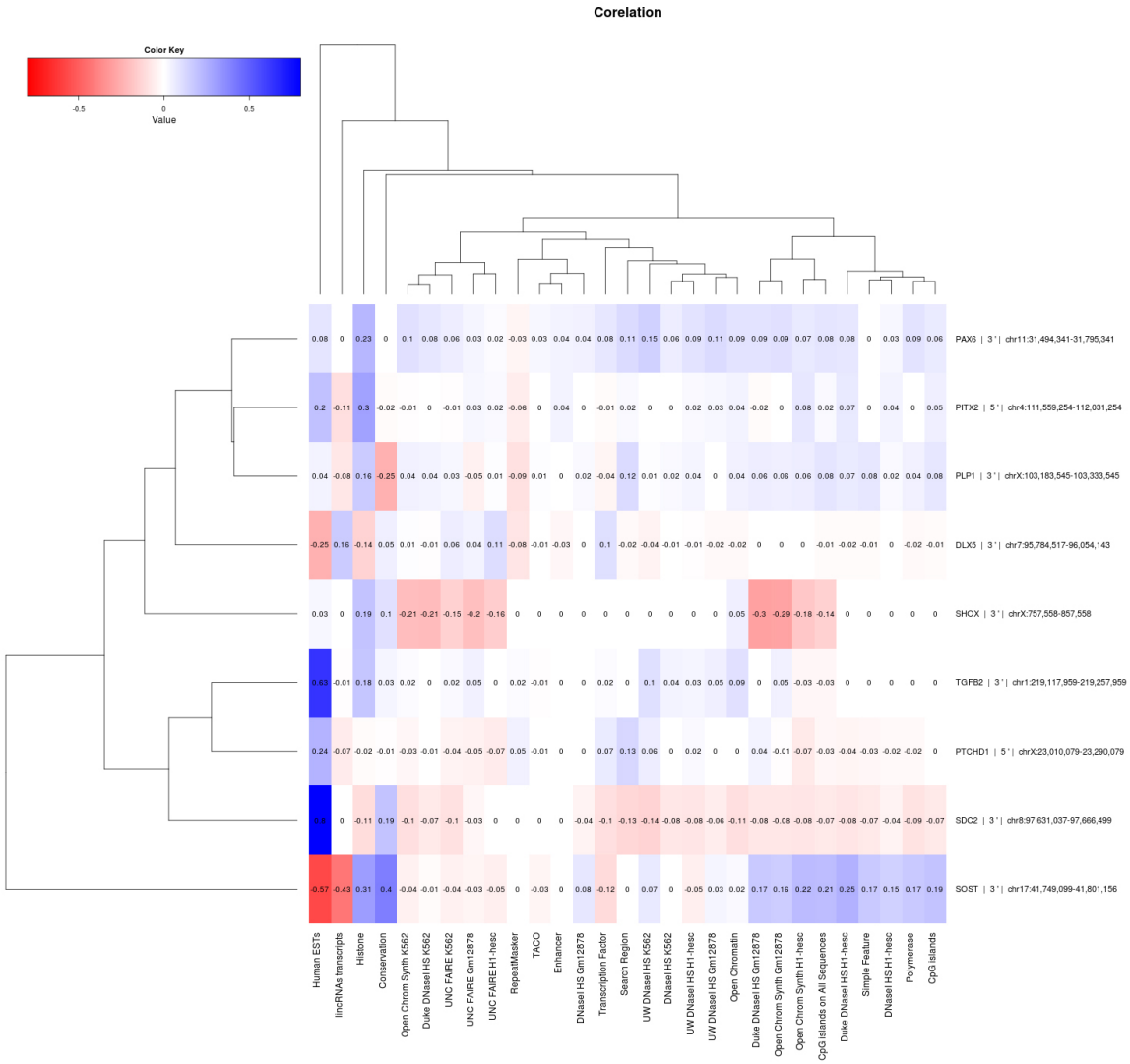

Figure 3.3: Pearson correlation between coverage of deletions of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of selected known HI genes.

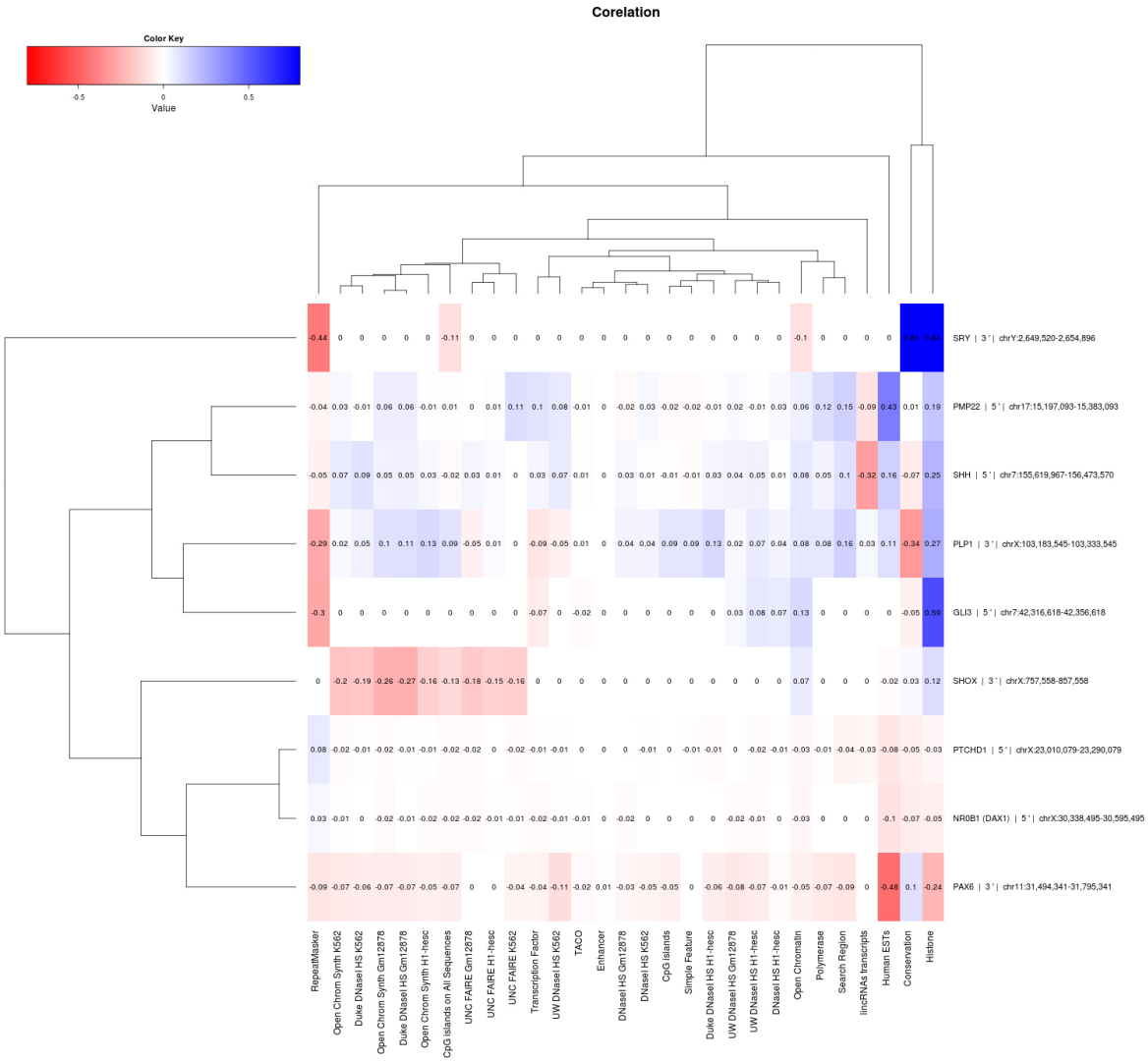

Figure 3.4: Pearson correlation between coverage of duplications of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of selected known HI genes.

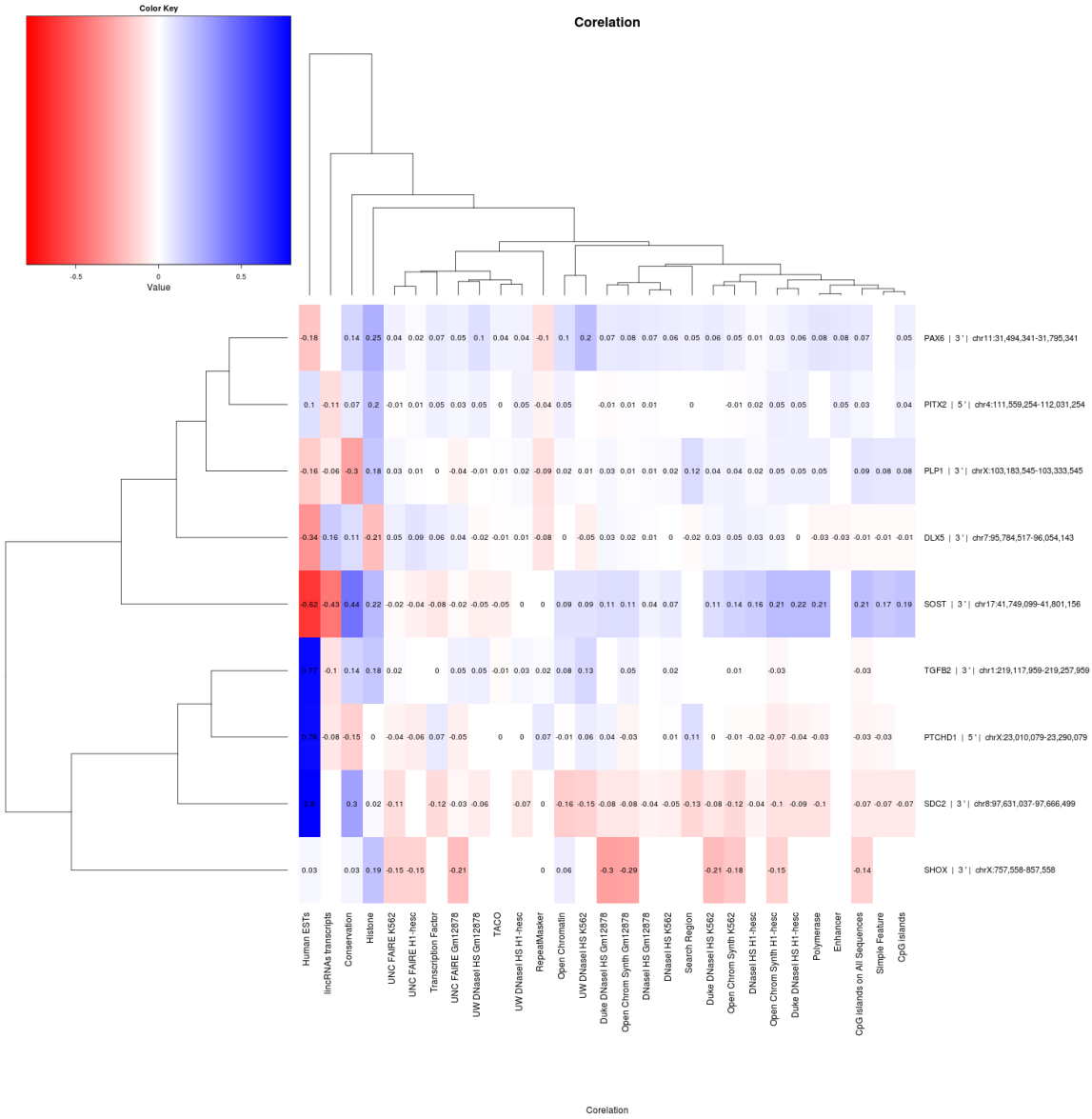

Figure 3.5: Spearman correlation between coverage of deletions of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of selected known HI genes.

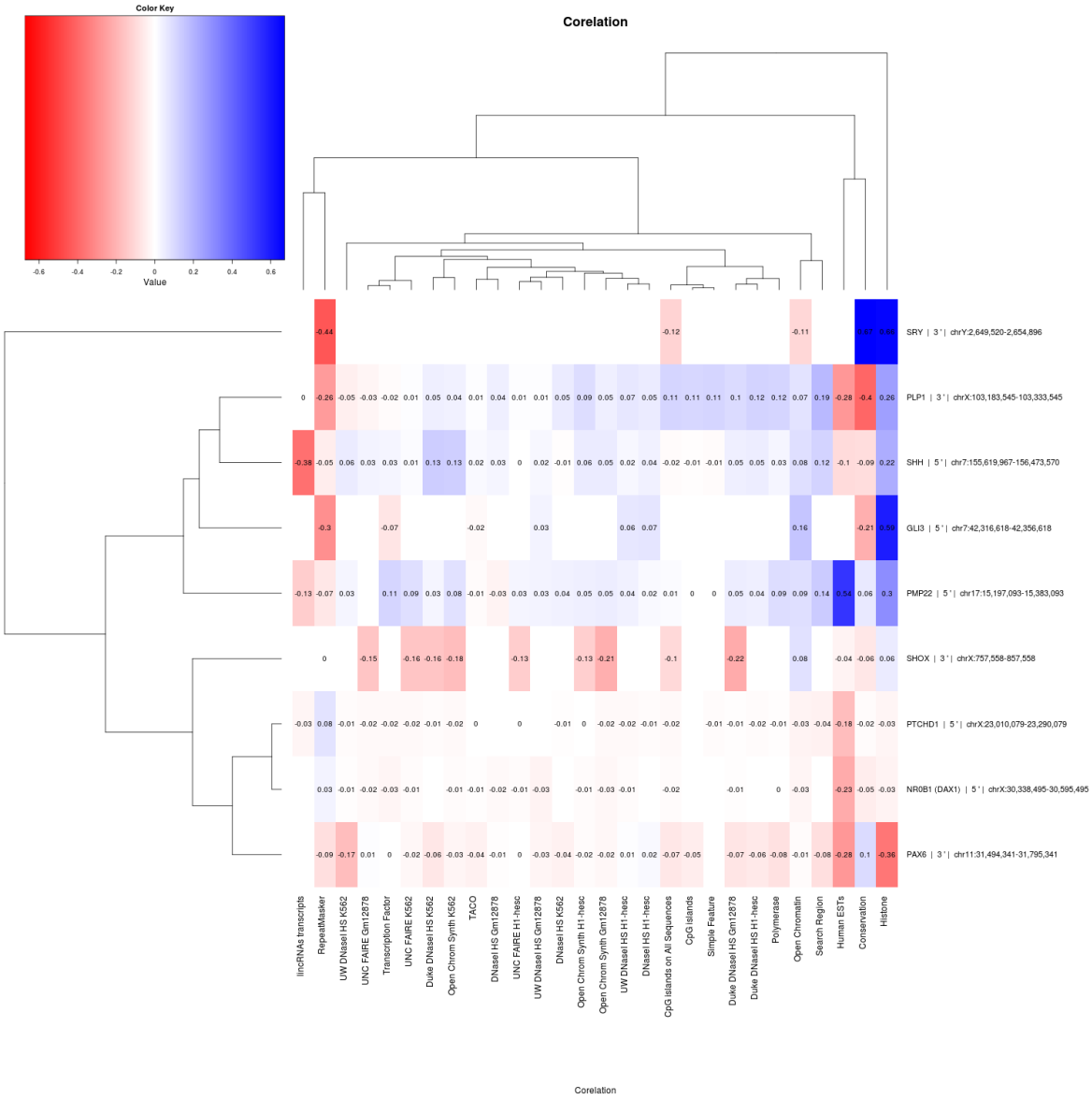

Figure 3.6: Spearman correlation between coverage of duplications of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of selected known HI genes.

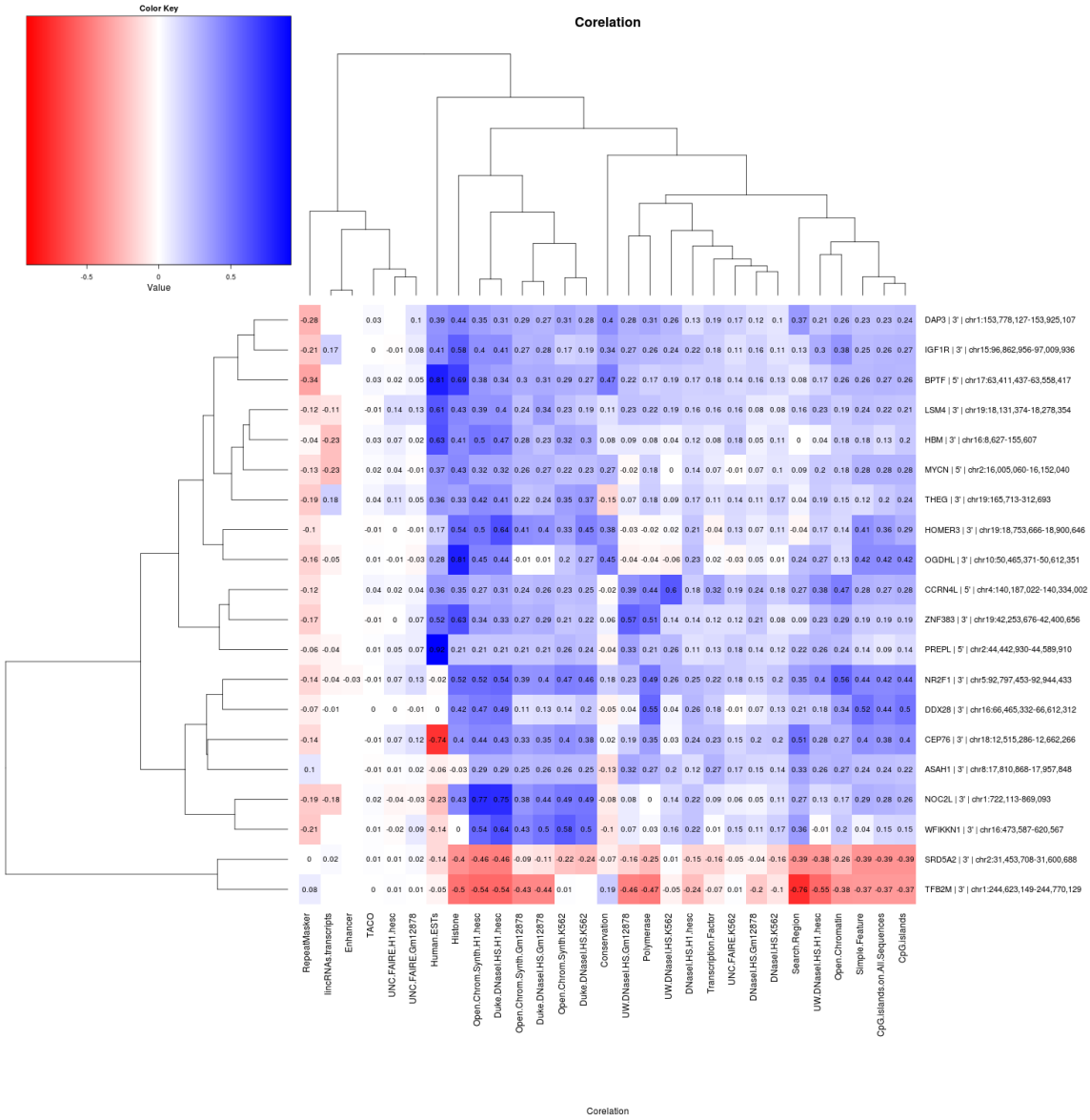

Figure 3.7: Pearson correlation between coverage of deletions of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of HI genes computationally predicted.

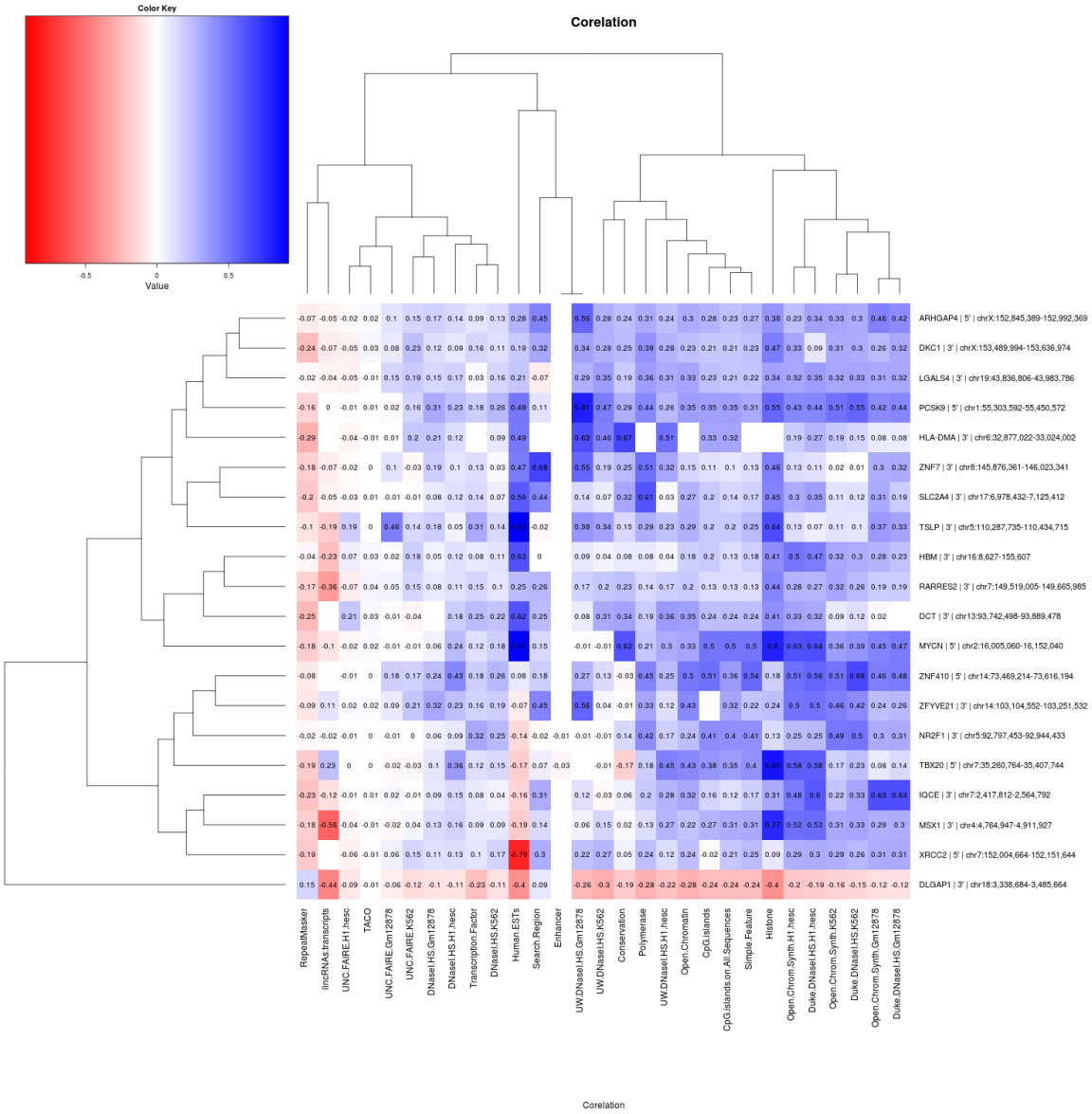

Figure 3.8: Pearson correlation between coverage of duplications of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of HI genes computationally predicted.

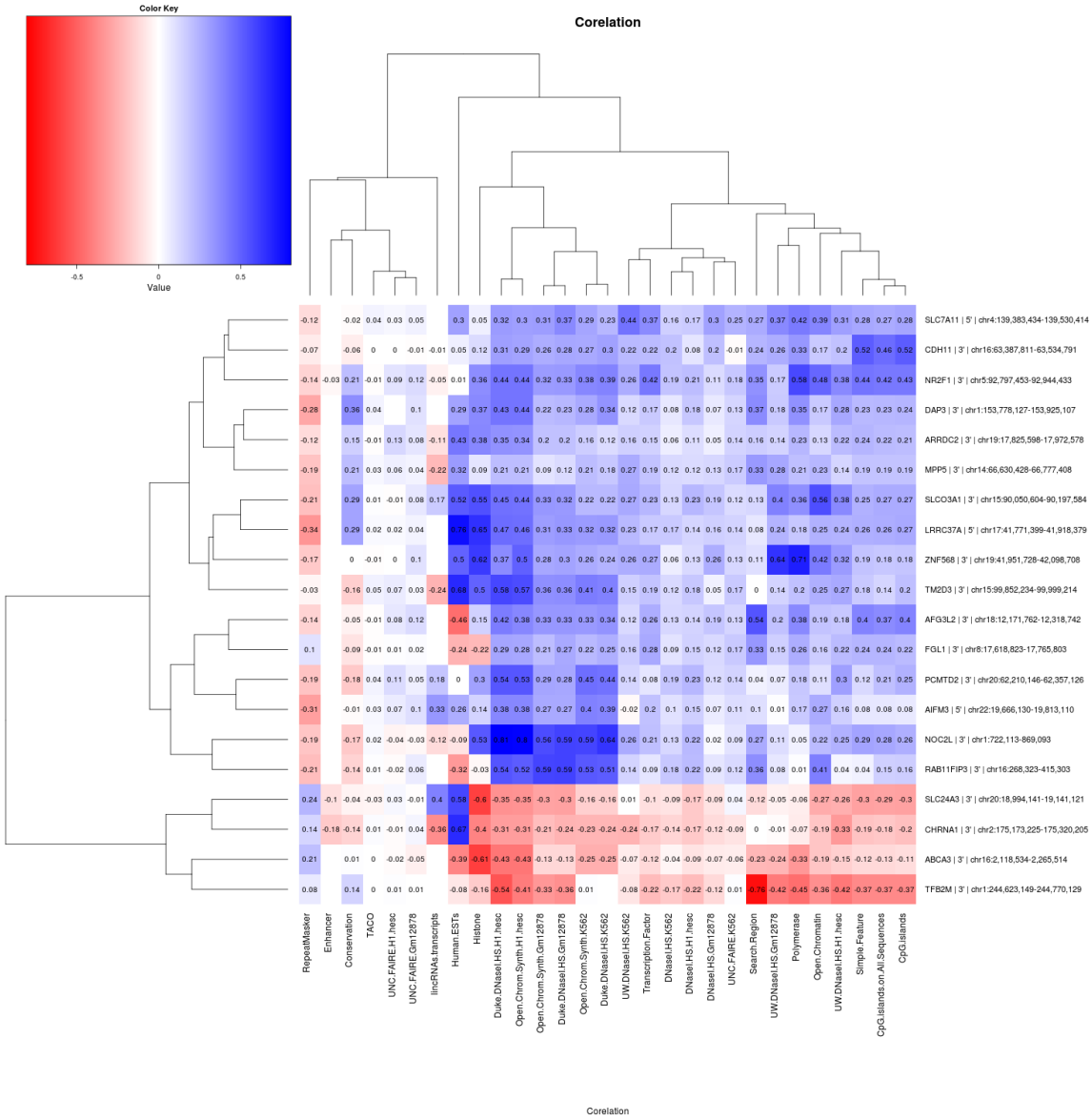

Figure 3.9: Spearman correlation between coverage of deletions of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of HI genes computationally predicted.

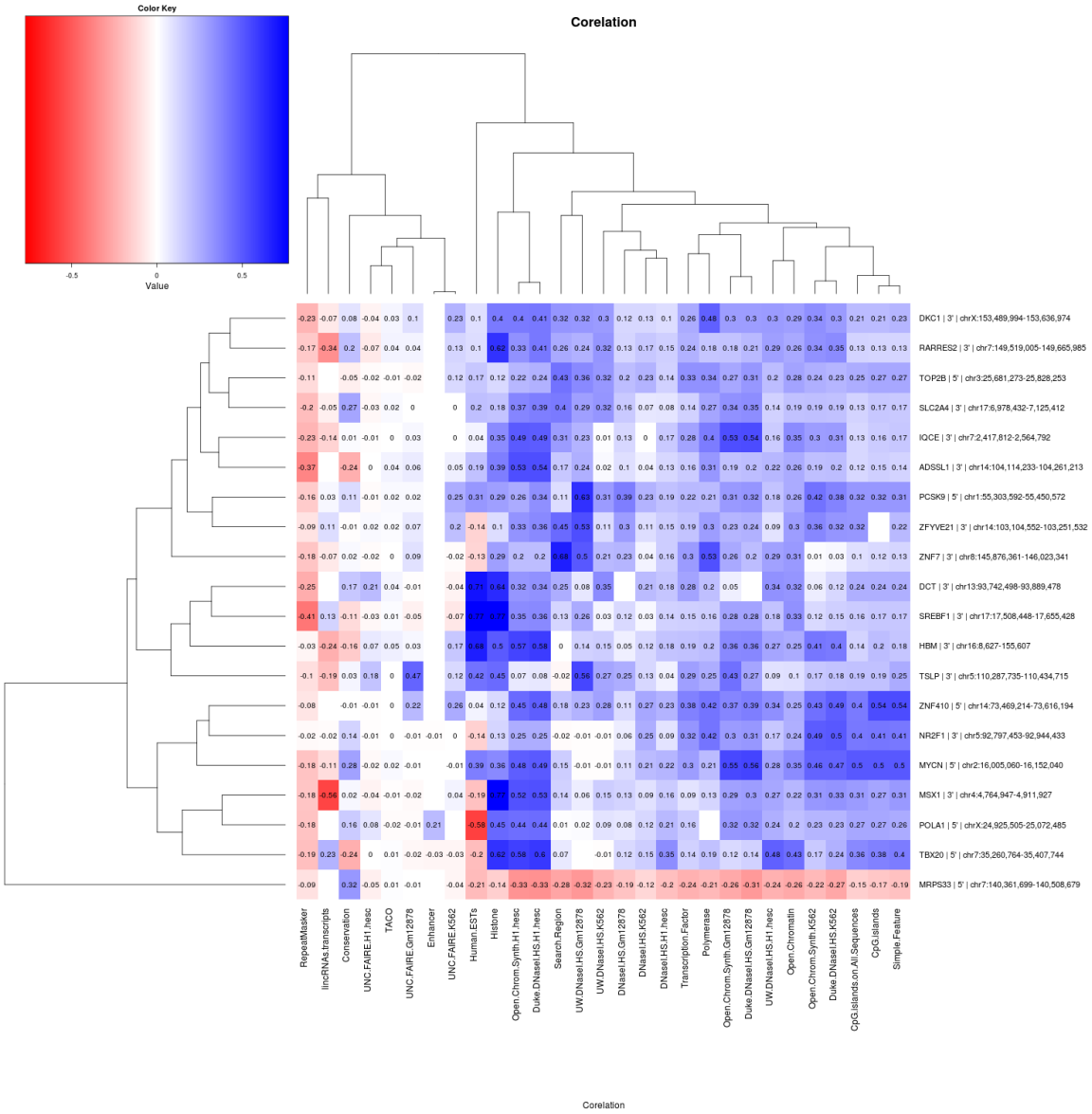

Figure 3.10: Spearman correlation between coverage of duplications of maximum length 1 Mb and a level or coverage of selected genomic features within potential regulatory regions of HI genes computationally predicted.

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